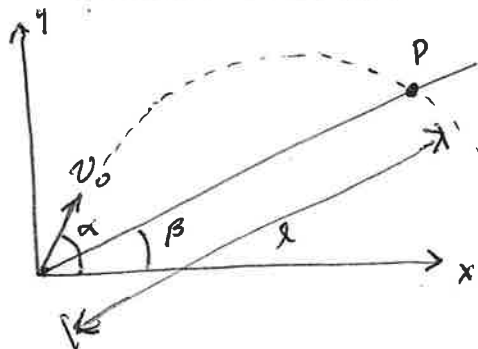


1)



$$\vec{F} = m \vec{a} \rightarrow m \ddot{x} = 0 \quad m \ddot{y} = -mg$$

$$\dot{x}(t=0) = v_0 \cos \alpha$$

$$\dot{y}(t=0) = v_0 \sin \alpha$$

 $t > 0$

$$\dot{x} = v_x = v_0 \cos \alpha$$

$$\dot{y} = v_y = v_0 \sin \alpha - gt$$

$$\int_0^t \frac{dx}{dt} = \int_0^t v_0 \cos \alpha dt \rightarrow x(t) = v_0 t \cos \alpha$$

$$\int_0^t \frac{dy}{dt} = \int_0^t (v_0 \sin \alpha - gt) dt \rightarrow y(t) = v_0 t \sin \alpha - \frac{gt^2}{2}$$

$$a) y(t') = l \sin \beta = v_0 t' \sin \alpha - \frac{gt'^2}{2}$$

$$x(t') = l \cos \beta = v_0 t' \cos \alpha$$

$$\Rightarrow t' = \frac{l \cos \beta}{v_0 \cos \alpha}$$

b) substitute $t' \rightarrow y(t')$ and solve for l

$$l \sin \beta = v_0 \left(\frac{l \cos \beta}{v_0 \cos \alpha} \right) \sin \alpha - \frac{1}{2} g \left(\frac{l^2 \cos^2 \beta}{v_0^2 \cos^2 \alpha} \right)$$

$$l = \left(\frac{\cos \beta}{\cos \alpha} \sin \alpha - \sin \beta \right) \cdot \frac{2 v_0^2 \cos^2 \alpha}{g \cos^2 \beta}$$

$$= \frac{2 v_0^2 \cos \alpha}{g \cos^2 \beta} \left(\cos \alpha \frac{\cos \beta \sin \alpha}{\cos \alpha} - \cos \alpha \sin \beta \right)$$

$$\Rightarrow l = \frac{2 v_0^2 \cos \alpha}{g \cos^2 \beta} \sin(\alpha - \beta)$$

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$c) l_{max} = \left. \frac{dl}{d\alpha} \right|_{\alpha_0} = 0$$

$$\begin{aligned} \left. \frac{dl}{d\alpha} \right|_{\alpha_0} &= \frac{2v_0^2}{g \cos^2 \beta} \left[-\sin \alpha \cdot \sin(\alpha - \beta) + \cos \alpha \cos(\alpha - \beta) \right] = 0 \\ &= \cos(\alpha + \alpha - \beta) = \cos(2\alpha - \beta) \\ \Rightarrow \cos(2\alpha_0 - \beta) &= 0 \Rightarrow 2\alpha_0 - \beta = \frac{\pi}{2} \end{aligned}$$

$$\boxed{\alpha_0 = \frac{\pi}{4} + \frac{\beta}{2}}$$

$$d) l_{max} = l(\alpha_0)$$

$$= \frac{2v_0^2}{g \cos^2 \beta} \cos\left(\frac{\pi}{4} + \frac{\beta}{2}\right) \cdot \underbrace{\sin\left(\frac{\pi}{4} + \frac{\beta}{2} - \beta\right)}_{\sin\left(\frac{\pi}{4} - \frac{\beta}{2}\right)}$$

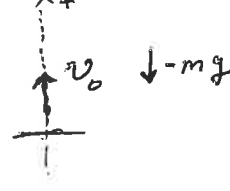
$$\cos\left(\frac{\pi}{4} + \frac{\beta}{2}\right) = \cos \frac{\pi}{4} \cos \frac{\beta}{2} - \sin \frac{\pi}{4} \sin \frac{\beta}{2}$$

$$\sin\left(\frac{\pi}{4} - \frac{\beta}{2}\right) = \sin \frac{\pi}{4} \cos \frac{\beta}{2} - \cos \frac{\pi}{4} \sin \frac{\beta}{2}$$

$$\begin{aligned} \Rightarrow \cos\left(\frac{\pi}{4} + \frac{\beta}{2}\right) \cdot \sin\left(\frac{\pi}{4} - \frac{\beta}{2}\right) &= \cos \frac{\pi}{4} \sin \frac{\pi}{4} \cos^2 \frac{\beta}{2} - \cos^2 \frac{\pi}{4} \cos \frac{\beta}{2} \sin \frac{\beta}{2} \\ &\quad - \sin^2 \frac{\pi}{4} \sin \frac{\beta}{2} \cos \frac{\beta}{2} + \sin \frac{\pi}{4} \cos \frac{\pi}{4} \sin^2 \frac{\beta}{2} \\ &= \underbrace{\sin \frac{\pi}{4} \cos \frac{\pi}{4}}_{\frac{1}{2}} - \sin \frac{\beta}{2} \cos \frac{\beta}{2} \underbrace{\left(\sin^2 \frac{\pi}{4} + \cos^2 \frac{\pi}{4} \right)}_{=1} \\ &= \frac{1}{2} - \frac{2 \sin \frac{\beta}{2} \cos \frac{\beta}{2}}{2} \end{aligned}$$

$$\Rightarrow l_{max} = \frac{2v_0^2}{g \cos^2 \beta} \cdot \frac{1}{2} (1 - \sin \beta) = \frac{1}{2} \left(1 - \underbrace{2 \sin \frac{\beta}{2} \cos \frac{\beta}{2}}_{\sin(2 \cdot \frac{\beta}{2})} \right)$$

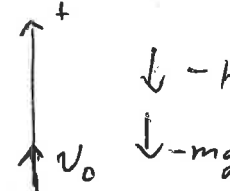
$$\boxed{l_{max} = \frac{v_0^2}{g(1 + \sin \beta)}}$$

2) a.  $F_{\text{net}} = -mg = m \frac{dv}{dt}$

$$\int_{v_0}^v dv = -\int_0^t g dt \rightarrow v(t) = v_0 - gt$$

$$v(t=t_{\text{max}}) = v_0 - gt_{\text{max}} = 0 \quad (\text{at max height})$$

$$\boxed{t_{\text{max}} = v_0/g}$$

b.  $F_{\text{net}} = -mg - kmv = m \frac{dv}{dt}$

$$dv = -(g + kv) dt$$

$$-\int_{v_0}^v \frac{dv}{(g + kv)} = \int_0^t dt \rightarrow \frac{1}{k} \ln(g + kv) \Big|_{v_0}^v = t$$

$$\ln(g + kv_0) - \ln(g + kv) = kt$$

$$\ln \left(\frac{g + kv_0}{g + kv} \right) = e^{kt} \rightarrow (g + kv) = (g + kv_0) e^{-kt}$$

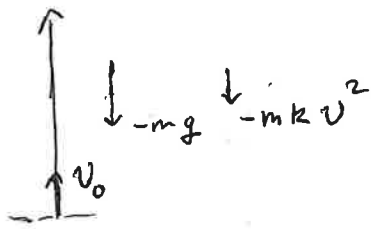
$$v(t) = \frac{(g + kv_0) e^{-kt}}{k} - \frac{g}{k} \Rightarrow v(t=t_{\text{max}}) = 0$$

$$\Rightarrow \ln e^{-kt_{\text{max}}} = \ln \frac{g}{(g + kv_0)}$$

$$-kt_{\text{max}} = \ln g / (g + kv_0)$$

$$\boxed{t_{\text{max}} = -\frac{1}{k} \ln \frac{g}{(g + kv_0)}}$$

3)



$$a) m \frac{dv}{dt} = -mg - mkv^2 \rightarrow \int_{v_0}^v \frac{dv}{-(g + kv^2)} = \int_0^t dt$$

$$-\frac{1}{g} \int \frac{dv}{(1 + \frac{k}{g}v^2)} \rightarrow \text{let } u^2 = \frac{k}{g}v^2, u = \sqrt{\frac{k}{g}}v, dv = \sqrt{\frac{g}{k}} du$$

$$-\frac{1}{g} \sqrt{\frac{g}{k}} \int \frac{du}{1+u^2} \rightarrow \tan^{-1}(u)$$

$$-\frac{1}{\sqrt{gk}} \tan^{-1}\left(\sqrt{\frac{k}{g}}v\right) \Big|_{v_0}^v = \int_0^t dt$$

$$\tan^{-1}\left(\sqrt{\frac{k}{g}}v\right) - \tan^{-1}\left(\sqrt{\frac{k}{g}}v_0\right) = -\sqrt{gk} \cdot t + \tan^{-1}\left(\sqrt{\frac{k}{g}}v_0\right)$$

$$\sqrt{\frac{k}{g}}v = \sqrt{\frac{g}{k}}t \tan\left(\tan^{-1}\sqrt{\frac{k}{g}}v_0 - \sqrt{gk}t\right)$$

$$V(t) = \sqrt{\frac{g}{k}} \tan\left(\tan^{-1}\sqrt{\frac{k}{g}}v_0 - \sqrt{gk}t\right)$$

Now, for terminal velocity $\Rightarrow$ if $F_r = mg$

$$\begin{array}{l} \uparrow F_{net} = 0 \\ \downarrow mg \end{array}$$

$$kv^2 = mg$$

$$v_t = \sqrt{\frac{g}{k}}$$

$$\Rightarrow V(t) = V_t \tan\left(\tan^{-1}\sqrt{\frac{V_0}{V_t}} - V_t k t\right)$$

$$\text{let } \alpha V_t = \tan^{-1}\left(\frac{V_0}{V_t}\right)$$

$$\Rightarrow V(t) = V_t \tan(\alpha V_t - kt V_t)$$

$$\boxed{V(t) = V_t \tan[V_t(\alpha - kt)]}$$

3) part b Find $y_{\max} = h$, let $V(t) = \frac{dy}{dt}$

$$\int_{y=0}^{y_{\max}} dy = V_t \int_0^{t^*} \tan[V_t(\alpha - kt)] dt, \quad t^* \rightarrow \text{time to reach } y_{\max}$$

let $U = V_t(\alpha - kt)$, $dU = -kV_t dt$

$$-\frac{V_t}{V_t k} \int \tan U dU = -\frac{1}{k} \left(-\ln [\cos(V_t(\alpha - kt))] \right) \Big|_0^{t^*}$$

determine t^*

$V(t^*) = 0$ at $y = y_{\max}$ (from part a)

$$V(t^*) = V_t \tan[V_t(\alpha - kt^*)] = 0$$

$$\Rightarrow V_t(\alpha - kt^*) = 0 \rightarrow V_t \alpha = V_t k t^*$$

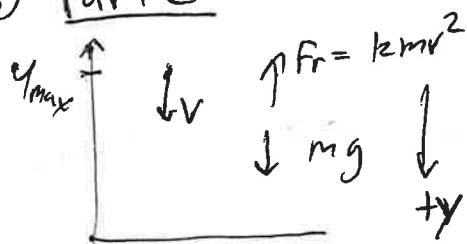
$$\boxed{t^* = \frac{\alpha}{k}} \rightarrow \text{Substitute into upper limit}$$

$$\begin{aligned} \Rightarrow y_{\max} &= +\frac{1}{k} \left[\ln \cos(V_t(\alpha - kt)) \right]_{t=0}^{t=\frac{\alpha}{k}} \\ &= \frac{1}{k} \left[\ln \left[\cos V_t \left(\alpha - k \frac{\alpha}{k} \right) \right] - \ln \cos(V_t \alpha) \right] \end{aligned}$$

$$\begin{aligned} &= 1 \\ \ln(1) &= 0 \end{aligned}$$

$$\boxed{y_{\max} = \frac{1}{k} \ln \frac{1}{\cos(\alpha V_t)}} \rightarrow \boxed{y_{\max} = \frac{1}{k} \ln \sec(\alpha V_t)}$$

3) Part C



$$t=0 \quad V(0)=0$$

$$mg - k m v^2 = m \frac{dv}{dy} \frac{dy}{dt}$$

$$g - k v^2 = \frac{dv}{dy} \cdot v$$

$$\int_0^y dy = \int_0^v \frac{V dV}{(g - kV^2)} \rightarrow \text{let } u = g - kV^2$$

$$du = -2kV dV$$

$$y(V) = -\frac{1}{2k} \int_{u=g}^u \frac{du}{u} = -\frac{1}{2k} \left[\ln \left(\frac{g - kV^2}{g} \right) \right]$$

$$k y = -\frac{1}{2} \ln \left(1 - \frac{kV^2}{g} \right) = \ln \left(1 - \frac{kV^2}{g} \right)^{-1/2}$$

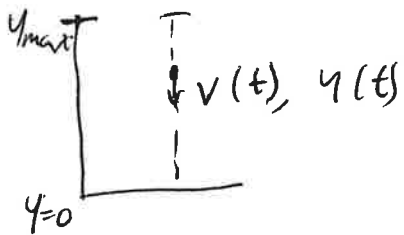
$$= \ln \frac{\sqrt{g} \sqrt{k}}{\sqrt{g - kV^2}} = \ln \frac{V_t}{\sqrt{V_t^2 - V^2}}$$

Recall $V_t^2 = \frac{g}{k}$

$$y = \frac{1}{k} \ln \left[\frac{V_t}{\sqrt{V_t^2 - V^2}} \right]$$

3) Part d

Recall from part b: $y_{\max} = \frac{1}{k} \ln \sec(\alpha V_t)$



Part c: $y(v) = \frac{1}{k} \ln \frac{V_t}{\sqrt{V_t^2 - v^2}}$

Set $y(v) = y_{\max}$ and solve for V
 velocity of particle once it returns to its initial position

$$\frac{1}{k} \ln \frac{V_t}{\sqrt{V_t^2 - v^2}} = \frac{1}{k} \ln \sec(\alpha V_t)$$

$$\frac{V_t}{\sqrt{V_t^2 - v^2}} = \sec(\alpha V_t) \rightarrow V_t \cos(\alpha V_t) = \sqrt{V_t^2 - v^2}$$

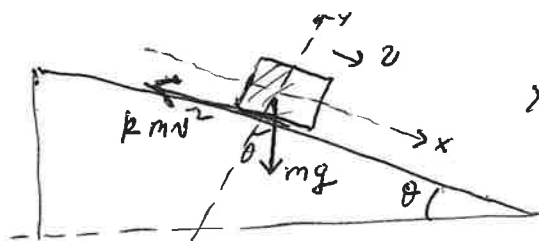
$$\Rightarrow \cos(\tan^{-1} \frac{V_0}{V_t}) \rightarrow \begin{array}{c} \sqrt{V_0^2 + V_t^2} \\ \theta = \alpha V_t \\ V_t \end{array} \quad \tan^{-1}(\frac{V_0}{V_t})$$

$$\Rightarrow \cos \alpha = \frac{V_t}{\sqrt{V_0^2 + V_t^2}} \Leftrightarrow \left(\frac{V_t}{\sqrt{V_0^2 + V_t^2}} \right)^2 = \left(\frac{\sqrt{V_t^2 - v^2}}{V_t} \right)^2 \Rightarrow V_t^2 - v^2 = \frac{V_t^4}{V_0^2 + V_t^2}$$

$$\Rightarrow v^2 = V_t^2 - \frac{V_t^4}{V_0^2 + V_t^2} = \frac{V_t^2 (V_0^2 + V_t^2) - V_t^4}{V_0^2 + V_t^2}$$

$$v^2 = \frac{V_t^2 V_0^2 + \cancel{V_t^4} - \cancel{V_t^4}}{V_0^2 + V_t^2} \Rightarrow \boxed{v = \frac{V_0 V_t}{\sqrt{V_0^2 + V_t^2}}}$$

4)



$$x: \quad \frac{dV}{dt} = g \sin \theta - k v^2$$

Init. conditions

$$t=0 \quad V(t=0)=0, \quad x(t=0)=0$$

$$\int_0^t dt = \int_0^v \frac{dv}{(g \sin \theta - k v^2)} = \frac{1}{g \sin \theta} \int_0^v \frac{dv}{\left(1 - \frac{k v^2}{g \sin \theta}\right)} \quad \text{let } a \equiv \frac{k}{g \sin \theta}$$

$$= \frac{a}{k} \int_0^v \frac{dv}{(1 - a v^2)} = \frac{a}{k} \left. \frac{\tanh^{-1}(\sqrt{a} v)}{\sqrt{a}} \right|_{v=0}^v$$

$$t = \frac{a}{k \sqrt{a}} \left[\tanh^{-1} \sqrt{a} v \right]$$

$$t = \frac{\sqrt{a}}{k} \tanh^{-1} \sqrt{a} v \rightarrow \tanh\left(\frac{k t}{\sqrt{a}}\right) = \sqrt{a} v$$

$$V(t) = \frac{dx}{dt} = \frac{1}{\sqrt{a}} \tanh\left(\frac{k t}{\sqrt{a}}\right) \rightarrow \int dx = \frac{1}{\sqrt{a}} \int_0^t \tanh\left(\frac{k t}{\sqrt{a}}\right) dt$$

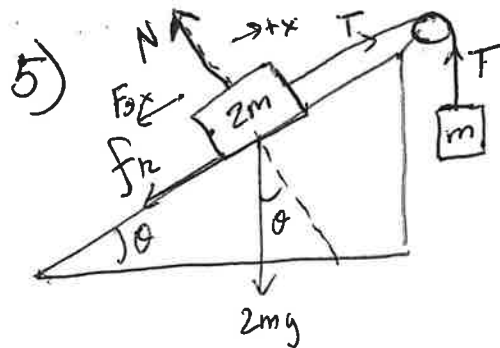
$$\Rightarrow \int_0^d dx = \frac{1}{\sqrt{a}} \left. \frac{\ln \cosh\left(\frac{k t}{\sqrt{a}}\right)}{\left(\frac{k}{\sqrt{a}}\right)} \right|_{t=0}^t$$

$$d = \frac{1}{k} \ln \cosh \frac{k t}{\sqrt{a}}$$

$$e^{k d} = \cosh \frac{k t}{\sqrt{a}} \rightarrow \cosh \frac{k t}{\sqrt{a}} = e^{k d}$$

$$\Rightarrow \frac{k t}{\sqrt{a}} = \cosh^{-1} e^{k d} \rightarrow t = \frac{\sqrt{a}}{k} \cosh^{-1} e^{k d}$$

$$t = \frac{1}{k} \frac{k^{1/2}}{(g \sin \theta)^{1/2}} \cosh^{-1}(e^{k d}) \Rightarrow \boxed{t = \frac{\cosh^{-1}(e^{k d})}{\sqrt{k g \sin \theta}}}$$



$$\sum F_x = ma_x$$

$$\sum F_y = Tmg - T = ma_y$$

$$N = 2mg \cos \theta$$

$$f_k = \mu_k N = \mu_k 2mg \cos \theta$$

$$F_{gx} = 2mg \sin \theta$$

for a particular $\theta = \theta_0 \rightarrow a_x = 0 \quad a_y = 0$ (const. speed)

$$\Rightarrow \sum F_y = mg - T = 0 \Rightarrow T = mg$$

$$\sum F_x = T - f_k - F_{gx} = 0$$

$$= mg - \mu_k 2mg \cos \theta - 2mg \sin \theta = 0$$

$$\Rightarrow 1 - 2\mu_k \cos \theta - 2 \sin \theta = 0$$

To solve for θ , need to express $\cos \theta$ in terms of $\sin \theta$
and re-group to solve for θ (need to use quadratic formula)

$$\Rightarrow 1 - 2\mu_k (1 - \sin^2 \theta)^{1/2} - 2 \sin \theta = 0$$

$$[1 - 2 \sin \theta]^2 = [2\mu_k (1 - \sin^2 \theta)^{1/2}]^2$$

$$1 + 4 \sin^2 \theta - 4 \sin \theta = 4\mu_k^2 (1 - \sin^2 \theta)$$

$$1 + 4 \sin^2 \theta - 4 \sin \theta = 4\mu_k^2 - 4\mu_k^2 \sin^2 \theta$$

$$4 \sin^2 \theta + 4\mu_k^2 \sin^2 \theta - 4 \sin \theta + 1 - 4\mu_k^2 = 0$$

$$4 \sin^2 \theta (1 + \mu_k^2) - 4 \sin \theta + 1 - 4\mu_k^2 = 0$$

$$\text{let } a \equiv 4(1 + \mu_k^2), \quad b \equiv -4, \quad c \equiv 1 - 4\mu_k^2$$

$$\Rightarrow a \sin^2 \theta + b \sin \theta + c = 0 \Rightarrow \text{quadratic formula}$$

$$\sin \theta = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

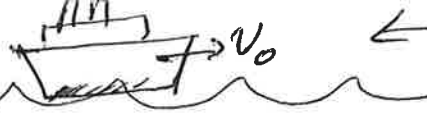
5) continued.

$$\Rightarrow \sin \theta = \frac{+4 \pm \sqrt{16 - 16(1 + \mu_k^2)(1 - 4\mu_k^2)}}{8(1 + \mu_k^2)}$$

$$= 4 \pm \frac{\sqrt{16(1 - (1 + \mu_k^2)(1 - 4\mu_k^2))}}{8(1 + \mu_k^2)}$$
$$4 \sqrt{1 - (1 - 4\mu_k^2 + \mu_k^2 - 4\mu_k^4)}$$
$$\sqrt{3\mu_k^2 + 4\mu_k^4}$$

$$\Rightarrow \sin \theta = 4 \pm 4\sqrt{\mu_k^2(3 + 4\mu_k^2)} / 8(1 + \mu_k^2)$$

$$\sin \theta_0 = \frac{1 \pm \mu_k \sqrt{3 + 4\mu_k^2}}{2(1 + \mu_k^2)}$$

6)  $F_r = -\alpha e^{\beta V}$

Part a Find $V(t)$

$$F_r = m \frac{dV}{dt} \Rightarrow -\alpha e^{\beta V} = m \frac{dV}{dt} \Rightarrow \int_{V_0}^V e^{-\beta V} dV = -\frac{\alpha}{m} \int_0^t dt$$

$$\Rightarrow -\frac{1}{\beta} \left[e^{-\beta V} \right]_{V_0}^V = -\frac{\alpha}{m} t$$

$$-\frac{1}{\beta} [e^{-\beta V} - e^{-\beta V_0}] = -\frac{\alpha t}{m} \Rightarrow \ln(e) = \ln(e^{-\beta V_0} + \frac{\alpha \beta}{m} t)$$

$$-\beta V = \ln(e^{-\beta V_0} + \frac{\alpha \beta}{m} t)$$

$$\Rightarrow V(t) = -\frac{1}{\beta} \ln\left(\frac{\alpha \beta}{m} t + e^{-\beta V_0}\right)$$

Part b Find the time for the boat to stop

$$V(t^*) = 0, \quad t^*$$

$$V(t^*) = -\frac{1}{\beta} \ln\left(\frac{\alpha \beta}{m} t^* + e^{-\beta V_0}\right) = 0$$

$$\Rightarrow \frac{\alpha \beta}{m} t^* = 1 - e^{-\beta V_0}$$

$$t^* = \frac{m}{\alpha \beta} - \frac{m}{\alpha \beta} e^{-\beta V_0} = \frac{m}{\alpha \beta} (1 - e^{-\beta V_0})$$

6) Part c Find the distance for boat to stop
 $X(t^*) \rightarrow$ dist. to stop, let $V(t) = \frac{dx}{dt}$

$$\rightarrow \int_0^{t^*} dx = \int_0^{t^*} V(t) dt$$

$$X = -\frac{1}{\beta} \int \ln \left[\frac{\alpha \beta}{m} t + e^{-\beta V_0} \right] dt \Leftrightarrow \int \ln(at+b)$$

where $a = \frac{\alpha \beta}{m}$
 $b = e^{-\beta V_0}$

~~$$= -\frac{1}{\beta} \int \ln \left[\frac{\alpha \beta}{m} \frac{m}{\alpha \beta} (1 - e^{-\beta V_0}) + e^{-\beta V_0} \right] dt$$~~
~~$$= -\frac{1}{\beta} \int \ln \left[\dots \right] dt$$~~

$$X(t) = -\frac{1}{\beta} \int_0^{t^*} \ln(at+b) dt, \quad a = \frac{\alpha \beta}{m} \quad b = e^{-\beta V_0}$$

$$\left. \frac{(at+b)}{a} \ln(at+b) - t \right|_0^{t^*}$$

$$X(t^*) = -\frac{1}{\beta} \left[\frac{(at^*+b)}{a} \ln(at^*+b) - t^* - \frac{b}{a} \ln(b) \right]$$

Now, Recall $at^*+b = \frac{\alpha \beta}{m} \frac{m}{\alpha \beta} (1 - e^{-\beta V_0}) + e^{-\beta V_0}$

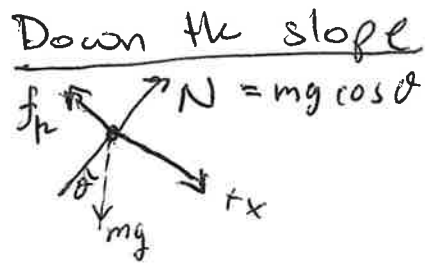
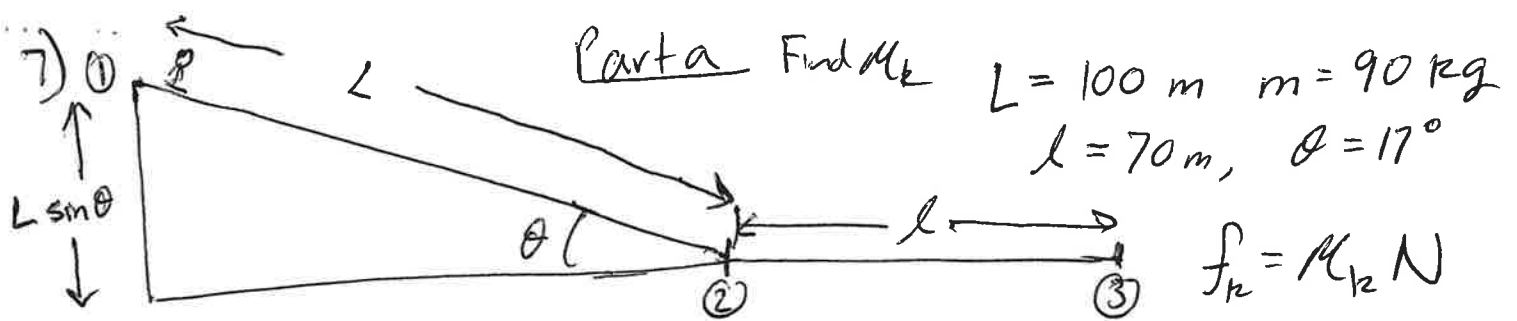
$$\underline{at^*+b = 1}$$

$$X(t^*) = -\frac{1}{\beta} \left[\frac{1}{a} \ln(1) - t^* - \frac{b}{a} \ln(b) \right]$$

$\downarrow$
0

$$= -\frac{1}{\beta} \left[-t^* - \frac{b}{a} \ln b \right] = +\frac{1}{\beta} \left[\frac{m}{\alpha \beta} (1 - e^{-\beta V_0}) + \frac{m}{\alpha \beta} e^{-\beta V_0} \ln(e^{-\beta V_0}) \right]$$

$$= \frac{m}{\alpha \beta^2} \left[1 - e^{-\beta V_0} + e^{-\beta V_0} (-\beta V_0) \right] = \boxed{\frac{m}{\alpha \beta^2} \left[1 - e^{-\beta V_0} (1 + \beta V_0) \right]}$$



$U_1 + K_1 = U_2 + K_2 + W_{1 \rightarrow 2}$ (work by friction)

$$mg L \sin \theta = \frac{1}{2} m v_2^2 + \mu_k mg \cos \theta L \quad (1)$$

Level snow

$$\frac{1}{2} m v_2^2 = W_{2 \rightarrow 3}, \quad W_{2 \rightarrow 3} = f_k l = \mu_k mg \cos \theta l$$

$$\frac{1}{2} m v_2^2 = \mu_k mg \cos \theta l \quad (2)$$

Plug (2) into (1): $\mu_k mg \cos \theta l + \mu_k mg \cos \theta L = mg L \sin \theta$

$$\mu_k l \cos \theta (l + L) = L \sin \theta$$

$$\boxed{\mu_k = \frac{L}{(L+l)} \tan \theta} \quad \text{or}$$

$$\frac{L \sin \theta}{L \cos \theta + l \cos \theta} \quad \cos \theta = 1 \quad (\text{at level snow, } \theta = 0)$$

Part b : from previous part

$$V_2 = \sqrt{\frac{2}{m} \mu_k mg \cos \theta l} \Rightarrow \boxed{V_2 = \sqrt{2 \mu_k g \cos \theta l}}$$

at level snow $\theta = 0$

or
$$\boxed{V_2 = \sqrt{2 \mu_k g l}}$$

$$8) U(x) = U_0 \left[2\left(\frac{x}{a}\right)^2 - \left(\frac{x}{a}\right)^4 \right]$$

$$a) F = -\nabla U = -\frac{dU}{dx}, \quad -\frac{dU}{dx} = -U_0 \left(4\left(\frac{x}{a}\right) \frac{1}{a} - 4\left(\frac{x}{a}\right)^3 \frac{1}{a} \right)$$

$$F = -4U_0 \left(\frac{x}{a^2} - \frac{x^3}{a^4} \right)$$

$$b) \frac{dU}{dx} = 0 \quad (\text{eq. points, where } F=0) \quad \text{eq. points}$$

$$\Rightarrow -4U_0 x \left(\frac{1}{a^2} - \frac{x^2}{a^4} \right) = 0 \Rightarrow$$

$$\begin{aligned} x_1 &= 0 \\ x_2 &= a \\ x_3 &= -a \end{aligned}$$

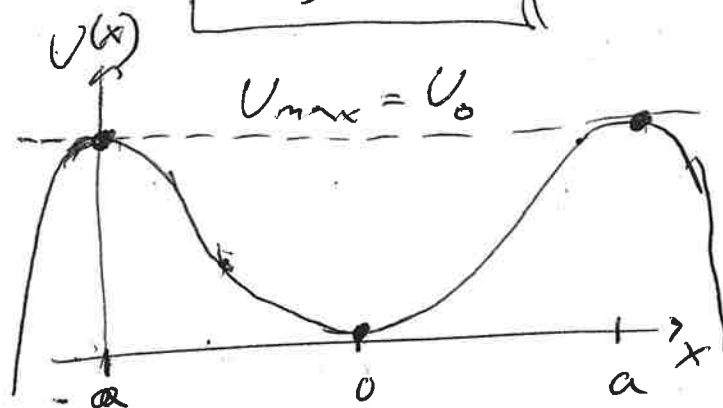
$$\left. \frac{d^2U}{dx^2} \right\} \begin{aligned} &> 0 \rightarrow \text{stable eq.} \\ &< 0 \rightarrow \text{unstable eq.} \end{aligned}$$

$$\frac{d^2U}{dx^2} = 4U_0 \left(\frac{1}{a^2} - \frac{3x^2}{a^4} \right)$$

$$\left. \frac{d^2U}{dx^2} \right|_{x_1=0} = \frac{4U_0}{a^2} > 0 \quad (\text{stable eq. at } x_1=0)$$

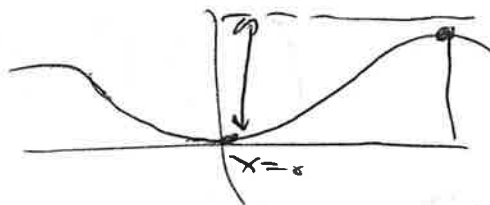
$$\left. \frac{d^2U}{dx^2} \right|_{x_2=a} = 4U_0 \left(\frac{a^2}{a^4} - \frac{3a^3}{a^4} \right) < 0 \quad (\text{unstable eq. at } x_2=a)$$

$$\left. \frac{d^2U}{dx^2} \right|_{x_3=-a} = 4U_0 \left(\frac{1}{a^2} - \frac{3a^2}{a^4} \right) < 0 \quad (\text{unstable eq. at } x_3=-a)$$



8) Part c

At the origin, $U(x=0)=0$



To be able to escape,

the minimum K.E. must be such that it is able

to overcome the potential barrier $U(x=\pm a)=U_0$

$$\Rightarrow \frac{1}{2} m V_{\min}^2 = U_0 \rightarrow \boxed{V_{\min} = \sqrt{\frac{2U_0}{m}}}$$

Part d

$$t=0 \quad X(0)=0 \quad V(t=0)=V_{\min}$$

$$\text{Now, } E_{\text{tot}} = \frac{1}{2} m V_{\min}^2 = U_0 \rightarrow \text{const.}$$

$$\frac{1}{2} m V_{\min}^2 = \frac{1}{2} m V(x)^2 + U(x) \equiv E \quad (\text{const.})$$

$$\rightarrow V(x) = \sqrt{\frac{2}{m} \left(\frac{1}{2} m V_{\min}^2 - U(x) \right)} = \sqrt{\frac{2}{m} (U_0 - U(x))}$$

$$\text{Now } V(x) \frac{dx}{dt} = \sqrt{\frac{2}{m} (U_0 - U(x))}$$

$$\rightarrow \frac{dx}{dt} = \sqrt{\frac{2U_0}{ma^4} \left(a^4 - 2a^2x^2 + x^4 \right)} = \sqrt{\frac{2U_0}{ma^4} (a^2 - x^2)^2}$$

$$\int \frac{dx}{\sqrt{a^4 - 2a^2x^2 + x^4}} = \int dt \sqrt{\frac{2U_0}{ma^4}} \rightarrow \int_0^x \frac{dx}{\sqrt{(a^2 - x^2)^2}} = \sqrt{\frac{2U_0}{ma^4}} \cdot t$$

$$= \frac{1}{a} \tanh^{-1}\left(\frac{x}{a}\right) \Big|_0^x$$

8) Part d (continued...)

$$\frac{dx}{a} \tanh^{-1}\left(\frac{x}{a}\right) = \sqrt{\frac{2V_0}{m}} \frac{dx}{a^2} \cdot t$$

$$\frac{x}{a} = \tanh \sqrt{\frac{2V_0}{ma^2}} \cdot t$$

$$\boxed{X(t) = a \cdot \tanh(\alpha t),} \quad \alpha \equiv \sqrt{\frac{2V_0}{ma^2}}$$